

Hands on 4

✓ 1. Java Persistence API (JPA)

- **What it is:**
A *Java specification* (JSR 338) that defines a standard for object-relational mapping (ORM) and data persistence in Java applications.
 - **Key point:**
JPA is **just a specification** — it doesn't have any working implementation on its own.
 - **Purpose:**
It allows Java developers to interact with relational databases using Java objects (i.e., POJOs) without writing SQL.
 - **Examples of JPA providers:**
 - **Hibernate**
 - EclipseLink
 - OpenJPA
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✓ 2. Hibernate

- **What it is:**
A popular **ORM framework** for Java that provides a **working implementation of the JPA specification**.
- **Features:**
 - Converts Java objects to database tables (ORM)
 - Automatically handles CRUD operations using session and transaction objects
 - Supports caching, lazy loading, and custom queries via HQL (Hibernate Query Language)
- **Hibernate code style** (from your example):

```
Session session = factory.openSession();
```

```
Transaction tx = session.beginTransaction();
```

```
session.save(employee);
```

```
tx.commit();
```

✓ 3. Spring Data JPA

- **What it is:**

A **Spring-based abstraction** built on top of JPA and Hibernate.
It doesn't implement JPA itself but simplifies its usage.

- **Main benefits:**

- **Reduces boilerplate code** (you don't have to write DAO implementations)
- Provides **automatic CRUD methods** via interfaces like JpaRepository
- Integrates easily with **Spring Boot**

- **Spring Data JPA code style:**

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer> {}
```

```
@Autowired
```

```
private EmployeeRepository repository;
```

```
repository.save(employee); // No session, transaction handling needed
```

🔄 Comparison Summary:

Feature	JPA	Hibernate	Spring Data JPA
Type	Specification (API)	Framework (Implementation)	Framework (Abstraction over JPA/Hibernate)
Boilerplate Code	Medium	High	Low
Implements JPA?	No	Yes	No
Requires DAO Layer?	Yes	Yes	No (Uses JpaRepository)
Session/Tx Handling	Manual	Manual	Auto-managed by Spring